

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1219

North Catskill 138 kV
June 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1219. The Interconnection Customer proposes a 133.9 MW solar photovoltaic facility interconnecting at the North Catskill 138 kV substation in the West Virginia area. The request was received on June 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 297 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,035,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Valemount-Hurricane 138kV	Pence-Valemount 345kV line	139.3	282	11.6
Recycling Coors-Hustisford 138kV	Vlemount-Pence 345kV	126.1	182	13.4
Recycling Coors-Peaceable 138kV L	Pence-Valemount 345kV	116.1	209	13.4
Greenleaf-Peaceable 138kV	Valemount-Pence 345kV	115.7	214	13.4
W Barstow-Hustisford 138kV	Pence-Valemount 345kV	114.9	191	13.4
Greenleaf-Tackett Creek 138kV	Valemount-Pence 345kV Line	112.9	231	13.4
Tackett Creek-Red Maple 138kV	Hurricane-N adison 345 138kV line	109.4	219	26.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hurricane-Valemount 138kV	Rebuild line with higher-capacity conductor	\$2,362,000	100% (\$2,362,000)
Hustisford-Recycling Coors 138kV	Rebuild line with higher-capacity conductor	\$6,999,000	100% (\$6,999,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Recycling Coors-Peaceable 138kV L	Rebuild line with higher-capacity conductor	\$5,610,000	100% (\$5,610,000)
Greenleaf-Peaceable 138kV	Rebuild line with higher-capacity conductor	\$2,211,000	100% (\$2,211,000)
W Barstow-Hustisford 138kV	Rebuild line with higher-capacity conductor	\$2,372,000	100% (\$2,372,000)
Greenleaf-Tackett Creek 138kV	Rebuild line with higher-capacity conductor	\$6,306,000	100% (\$6,306,000)
Tackett Creek-Red Maple 138kV	Reconductor line	\$1,175,000	100% (\$1,175,000)
	Total allocated to project		\$27,035,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.